

ALEX LEVY

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EDUCATION

Virginia Tech – Blacksburg, VA

M.Eng. – Computer Science and Applications (4+1 Accelerated)

Expected May 2027

B.S. – Computer Science, Applied Discrete Mathematics

May 2026 • 3.97 GPA

- Coursework: Database Management Systems, Multiprocessor Programming, Computer Architecture, Graph Theory, Machine Learning, Programming Languages.

Montgomery College – Rockville, MD

May 2023

A.S. – Mathematics

- Dual Enrollment Degree - 95 college credits earned while in high school.

RELEVANT EXPERIENCE

Software Engineering Intern – eClinicalWorks, Westborough, MA

June 2026 - August 2026

- Built a modular Python pipeline to extract patient name, date of birth, and document type from variable-quality medical PDFs, combining PDF parsing, OCR fallback, classification, NER, LLM extraction, validation, and export.
- Benchmarked local CPU and A100 GPU pipelines across OCR/LLM backends; DeepSeekVL parsed nearly all samples at 8.7 s/page, while EasyOCR averaged 7.37 s/page, informing confidence-gated routing by document hardness.

Research Assistant – Virginia Tech, Blacksburg, VA

January 2026 - Present

- Developed a Scala automatic test-case minimization system for fuzzing distributed dataflow frameworks including Apache Spark, Flink, and Polars.
- Applied rule-based search to prune irrelevant dataflow components while preserving failure-inducing behavior, minimizing 9 confirmed bugs across 4 major frameworks.

Teaching Assistant – Virginia Tech, Blacksburg, VA

August 2025 – May 2026

- Provided detailed feedback on Python assignments and led office hours supporting 100+ students with debugging and core programming concepts.

PROJECTS

LevyCAS – Personal Project

Summer 2025

- Engineered a multivariate computer algebra system from scratch in Python, implementing a custom Pratt parser for symbolic expression parsing.
- Implemented polynomial GCD, symbolic differentiation/integration, algebraic simplification, and adaptive terminal graphing; deployed interactive demos with Textual, FastAPI, and a Vercel monorepo.

Spritz – Personal Project

Fall 2025

- Built a C++ ray-tracing engine from scratch and used it as an inverse graphics testbed to recover scene parameters, including camera pose and geometry, from rendered images.

LoX Robustness Study – Course Capstone, Virginia Tech

Spring 2026

- Developed an evaluation pipeline to study whether benign fine-tuning can weaken LoX (Low-Rank Extrapolation), using sequential LoRA/Unsloth fine-tuning of Llama-2-7B-Chat on GSM8K and BeaverTails.
- Measured safety degradation with AdvBench and Llama-Guard-3-8B, finding LoX reduced direct-attack ASR from 91% to 37% at 100 harmful samples, but 1,000 benign GSM8K samples raised ASR to 63% at 100 and 89% at 1,000 harmful samples.

Eagle Scout Project - Troop 457, Rockville, MD

Summer 2023

- Designed, planned, and led a 20+ volunteer team over three months to construct six 42 sq. ft. garden beds for a local community park.

SKILLS

- Languages: Python, C++, Scala, C, Java.
- Frameworks & Libraries: Spark, Flink, Polars, Dask, NumPy, SymPy, React, Next.js, TailwindCSS, Gradio, Textual.
- ML / AI: OCR, NER, LLMs, LoRA, Unsloth; Areas: Distributed Systems, Information Extraction, Symbolic Computation.

HONORS & LEADERSHIP

T.W. Hatcher Scholarship (2025); Ray A. Gaskins Scholarship (2024/2025); Clifford Beck Physics Award (2023); Eagle Scout (2023)